

Assignment No 1

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Subject : ADS Lab

Problem Statement: Assignment based on implementation of Artificial Neural Network (ANN) using TensorFlow/Keras for prediction problems, Exemplars: Student Performance Prediction, House Price Prediction, Employee Attrition Prediction

Implementation:

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# -----
# Load Dataset
# -----

housing = fetch_california_housing(as_frame=True)

df = housing.frame

print(df.head())
print(df.info())

# -----
# Feature and Target
# -----

X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]

# -----
# Train Test Split
# -----
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
# -----
# Feature Scaling
# -----
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# -----
# Build ANN Model
# -----
```

```
model = Sequential()
```

```
model.add(Dense(64,
                activation='relu',
                input_shape=(X_train.shape[1],)))
```

```
model.add(Dense(32,
                activation='relu'))
```

```
model.add(Dense(16,
                activation='relu'))
```

```
model.add(Dense(1,
                activation='linear'))
```

```
# -----
# Compile Model
# -----
```

```
model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['mae']
)
```

```
# -----
# Train Model
# -----
```

```
history = model.fit(
```

```

X_train,
y_train,
validation_split=0.2,
epochs=50,
batch_size=32,
verbose=1
)

# -----
# Evaluate Model
# -----

loss, mae = model.evaluate(X_test, y_test)

print("Test Loss :", loss)
print("Test MAE :", mae)

# -----
# Prediction
# -----

y_pred = model.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print("RMSE :", rmse)

# -----
# Plot Loss
# -----

plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.show()

```

Output:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585

```

2 7.2574    52.0 8.288136 1.073446    496.0 2.802260    37.85 -122.24
3.521
3 5.6431    52.0 5.817352 1.073059    558.0 2.547945    37.85 -122.25
3.413
4 3.8462    52.0 6.281853 1.081081    565.0 2.181467    37.85 -122.25
3.422

```

<class 'pandas.DataFrame'>

RangeIndex: 20640 entries, 0 to 20639

Data columns (total 9 columns):

```

#   Column      Non-Null Count  Dtype
---  -----
0  MedInc      20640 non-null float64
1  HouseAge    20640 non-null float64
2  AveRooms    20640 non-null float64
3  AveBedrms   20640 non-null float64
4  Population  20640 non-null float64
5  AveOccup    20640 non-null float64
6  Latitude    20640 non-null float64
7  Longitude   20640 non-null float64
8  MedHouseVal 20640 non-null float64

```

dtypes: float64(9)

memory usage: 1.4 MB

None

826/826 ————— 5s 6ms/step - loss: 0.4366

- mae: 0.4685 - val_loss: 0.4301 - val_mae: 0.4739

Epoch 3/50

826/826 ————— 4s 5ms/step - loss: 0.3876

- mae: 0.4469 - val_loss: 0.4093 - val_mae: 0.4604

Epoch 4/50

826/826 ————— 5s 6ms/step - loss: 0.3748

- mae: 0.4358 - val_loss: 0.3908 - val_mae: 0.4429

Epoch 5/50

826/826 ————— 4s 5ms/step - loss: 0.3577

- mae: 0.4262 - val_loss: 0.3833 - val_mae: 0.4456

Epoch 6/50

826/826 ————— 4s 4ms/step - loss: 0.3491

- mae: 0.4193 - val_loss: 0.3885 - val_mae: 0.4288

Epoch 7/50

826/826 ————— 4s 5ms/step - loss: 0.3379

- mae: 0.4109 - val_loss: 0.3612 - val_mae: 0.4149

Epoch 8/50

826/826 ————— 5s 6ms/step - loss: 0.3277

- mae: 0.4024 - val_loss: 0.3655 - val_mae: 0.4174

Epoch 9/50

826/826 ————— 5s 5ms/step - loss: 0.3201

- mae: 0.3970 - val_loss: 0.3441 - val_mae: 0.4006

Epoch 10/50

826/826 ————— 4s 5ms/step - loss: 0.3149

- mae: 0.3929 - val_loss: 0.3394 - val_mae: 0.3996

Epoch 11/50
826/826 ————— 4s 5ms/step - loss: 0.3122
- mae: 0.3903 - val_loss: 0.3351 - val_mae: 0.4088

Epoch 12/50
826/826 ————— 4s 5ms/step - loss: 0.3074
- mae: 0.3869 - val_loss: 0.3475 - val_mae: 0.3996

Epoch 13/50
826/826 ————— 4s 5ms/step - loss: 0.3107
- mae: 0.3865 - val_loss: 0.3348 - val_mae: 0.3972

Epoch 14/50
826/826 ————— 4s 5ms/step - loss: 0.2993
- mae: 0.3827 - val_loss: 0.3432 - val_mae: 0.4017

Epoch 15/50
826/826 ————— 6s 7ms/step - loss: 0.2994
- mae: 0.3813 - val_loss: 0.3246 - val_mae: 0.3892

Epoch 16/50
826/826 ————— 6s 7ms/step - loss: 0.2984
- mae: 0.3797 - val_loss: 0.3297 - val_mae: 0.4030

Epoch 17/50
826/826 ————— 3s 4ms/step - loss: 0.2943
- mae: 0.3783 - val_loss: 0.3313 - val_mae: 0.4017

Epoch 18/50
826/826 ————— 6s 7ms/step - loss: 0.3019
- mae: 0.3794 - val_loss: 0.3177 - val_mae: 0.3857

Epoch 19/50
826/826 ————— 6s 7ms/step - loss: 0.2906
- mae: 0.3746 - val_loss: 0.3215 - val_mae: 0.3854

Epoch 20/50
826/826 ————— 5s 6ms/step - loss: 0.2920
- mae: 0.3749 - val_loss: 0.3221 - val_mae: 0.3968

Epoch 21/50
826/826 ————— 6s 7ms/step - loss: 0.2861
- mae: 0.3720 - val_loss: 0.3282 - val_mae: 0.3854

Epoch 22/50
826/826 ————— 6s 7ms/step - loss: 0.2862
- mae: 0.3717 - val_loss: 0.3419 - val_mae: 0.3907

Epoch 23/50
826/826 ————— 6s 7ms/step - loss: 0.2951
- mae: 0.3726 - val_loss: 0.3109 - val_mae: 0.3801

Epoch 24/50
826/826 ————— 5s 6ms/step - loss: 0.2840
- mae: 0.3691 - val_loss: 0.3190 - val_mae: 0.3807

Epoch 25/50
826/826 ————— 6s 7ms/step - loss: 0.2846
- mae: 0.3689 - val_loss: 0.3104 - val_mae: 0.3799

Epoch 26/50
826/826 ————— 6s 7ms/step - loss: 0.2837
- mae: 0.3688 - val_loss: 0.3235 - val_mae: 0.3843

Epoch 27/50
826/826 ————— 6s 7ms/step - loss: 0.2872
- mae: 0.3702 - val_loss: 0.3132 - val_mae: 0.3913

Epoch 28/50
826/826 ————— 6s 7ms/step - loss: 0.2793
- mae: 0.3666 - val_loss: 0.3166 - val_mae: 0.3837

Epoch 29/50
826/826 ————— 6s 8ms/step - loss: 0.2795
- mae: 0.3663 - val_loss: 0.3137 - val_mae: 0.3840

Epoch 30/50
826/826 ————— 3s 4ms/step - loss: 0.2792
- mae: 0.3660 - val_loss: 0.3055 - val_mae: 0.3767

Epoch 31/50
826/826 ————— 6s 7ms/step - loss: 0.2773
- mae: 0.3642 - val_loss: 0.3124 - val_mae: 0.3754

Epoch 32/50
826/826 ————— 3s 4ms/step - loss: 0.2774
- mae: 0.3642 - val_loss: 0.3335 - val_mae: 0.4043

Epoch 33/50
826/826 ————— 6s 7ms/step - loss: 0.2806
- mae: 0.3644 - val_loss: 0.3174 - val_mae: 0.3911

Epoch 34/50
826/826 ————— 5s 6ms/step - loss: 0.2853
- mae: 0.3648 - val_loss: 0.3169 - val_mae: 0.3778

Epoch 35/50
826/826 ————— 6s 8ms/step - loss: 0.2797
- mae: 0.3639 - val_loss: 0.3059 - val_mae: 0.3837

Epoch 36/50
826/826 ————— 5s 6ms/step - loss: 0.2744
- mae: 0.3616 - val_loss: 0.3185 - val_mae: 0.3785

Epoch 37/50
826/826 ————— 6s 7ms/step - loss: 0.2763
- mae: 0.3626 - val_loss: 0.3169 - val_mae: 0.4066

Epoch 38/50
826/826 ————— 5s 7ms/step - loss: 0.2771
- mae: 0.3618 - val_loss: 0.3045 - val_mae: 0.3765

Epoch 39/50
826/826 ————— 6s 7ms/step - loss: 0.2742
- mae: 0.3616 - val_loss: 0.3145 - val_mae: 0.3745

Epoch 40/50
826/826 ————— 5s 6ms/step - loss: 0.2756
- mae: 0.3637 - val_loss: 0.3062 - val_mae: 0.3904

Epoch 41/50
826/826 ————— 6s 7ms/step - loss: 0.2723
- mae: 0.3602 - val_loss: 0.3099 - val_mae: 0.3726

Epoch 42/50
826/826 ————— 11s 7ms/step - loss:
0.2770 - mae: 0.3629 - val_loss: 0.3148 - val_mae: 0.3750

Epoch 43/50
826/826 ————— 5s 6ms/step - loss: 0.2824
- mae: 0.3609 - val_loss: 0.3157 - val_mae: 0.3942
Epoch 44/50
826/826 ————— 5s 5ms/step - loss: 0.2712
- mae: 0.3607 - val_loss: 0.3082 - val_mae: 0.3814
Epoch 45/50
826/826 ————— 6s 7ms/step - loss: 0.2715
- mae: 0.3601 - val_loss: 0.3072 - val_mae: 0.3873
Epoch 46/50
826/826 ————— 3s 4ms/step - loss: 0.2699
- mae: 0.3578 - val_loss: 0.3161 - val_mae: 0.3929
Epoch 47/50
826/826 ————— 5s 6ms/step - loss: 0.2706
- mae: 0.3588 - val_loss: 0.3082 - val_mae: 0.3933
Epoch 48/50
826/826 ————— 4s 4ms/step - loss: 0.2712
- mae: 0.3602 - val_loss: 0.3050 - val_mae: 0.3775
Epoch 49/50
826/826 ————— 4s 5ms/step - loss: 0.2715
- mae: 0.3596 - val_loss: 0.3249 - val_mae: 0.3910
Epoch 50/50
826/826 ————— 6s 7ms/step - loss: 0.2702
- mae: 0.3582 - val_loss: 0.3011 - val_mae: 0.3727
129/129 ————— 1s 4ms/step - loss: 0.2858
- mae: 0.3622
Test Loss : 0.2857688069343567
Test MAE : 0.3622063100337982
129/129 ————— 0s 2ms/step
RMSE : 0.5345733598384023

